Qatar-1b

R-0605

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_221204` · created 2026-07-25T05:26:22+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459917.65 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.056
Transit depth (Rp/Rs)^2	3.6 +/- 2.14 [%]
Orbital Inclination (inc)	88.03 +/- 2.88
Airmass coefficient 1 (a1)	1.01 +/- 0.66
Airmass coefficient 2 (a2)	-0.13 +/- 0.47
Scatter in the residuals of the lightcurve fit is	55.72 %
Best Comparison Star	#1 - [206, 130]
Optimal Aperture	1.33
Optimal Annulus	5.33
Transit Duration (day)	0.0818 +/- 0.0082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.056 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.0818 d vs 0.0692 d (deviation 1.54σ)
Mid-time O–C	5.0 min
Depth SNR	3.4
Residual scatter	55.72 %

Light curve

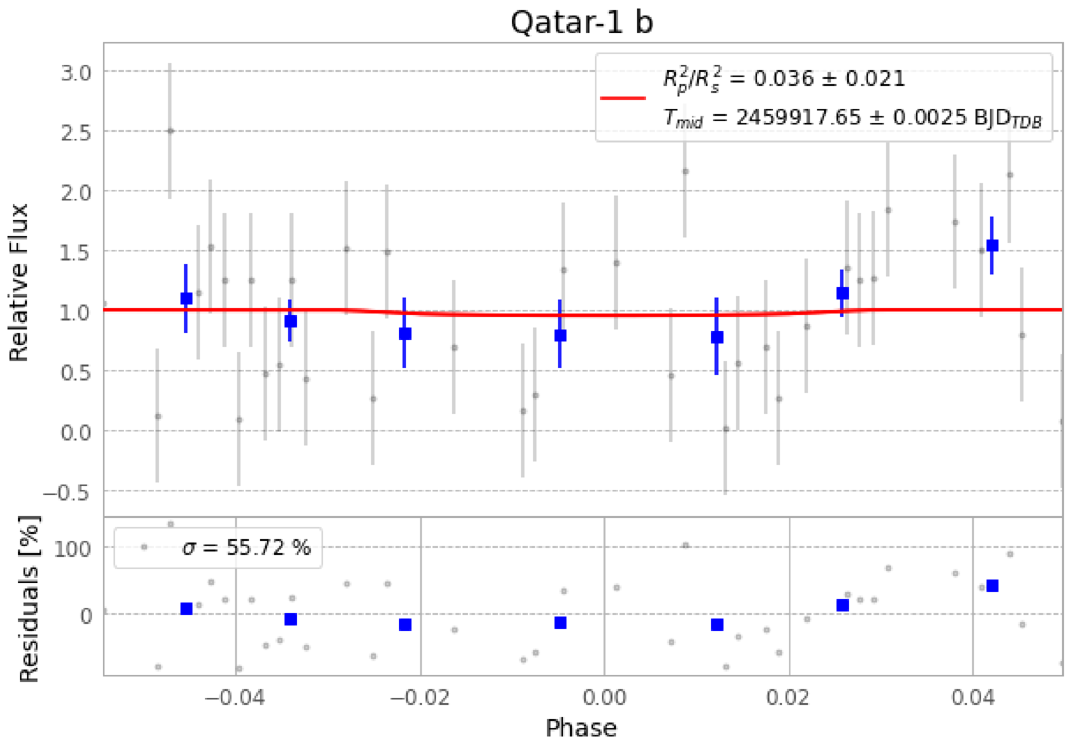

Figure 1 — FinalLightCurve_Qatar-1 b_2022-12-03.png (SHA-256 152d889c05292759…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [438, 271], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5e47957de9c208d6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

73 FITS frames (48 MB), `Qatar-1221204014215.FITS` → `Qatar-1221204051815.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-221204.log (892548924fad…), FinalLightCurve_Qatar-1 b_2022-12-03.png (152d889c0529…), FinalLightCurve_Qatar-1 b_2022-12-03.csv (41e2226ec905…)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)